Exam

D. Jason Koskinen
koskinen@nbi.ku.dk

Advanced Methods in Applied Statistics
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Photo by Howard Jackman

Info

- In submitting the solutions there is no need to rephrase the problem. For example, “Solution for 1a” is sufficient.
- The submission format for explanations and plots is a PDF file. Also, in a separate file (or files) include any and all software scripts used to establish your answers and/or produce plots.
- Working in groups or any communication about the problems is **prohibited**. Using the internet as a resource is encouraged, but soliciting any help is prohibited.
- Some questions have multiple parts. For full credit, all parts must be done.

Info

- The exam will be graded out of 10 possible points
  - It will count for 40% of the final course grade
- Submit all code used!! The software you write to complete the problem is **part** of the solution.
- The exam must be electronically submitted via the Digital Exam website.
  - For catastrophic submission failures you can email the exam submission to Jason
- Look through all problems in the exam. Some problems are easier than others.
- For any concerns, questions, or comments email Jason (koskinen@nbi.ku.dk)

Starting (0.5 pts.)

- On the first page of your write-up include your **full name**, **date**, **name of this course**, **UCPH ID**, and the **title of your exam submission**
- Also type out (please don't copy/paste) " I {*your name here*} expressly vow to uphold my scientific, academic, and moral integrity by working individually on this exam and soliciting no direct external help or assistance."
- Finding help/solutions online is fine. But, for example, posting to an online forum or generative AI model and receiving assistance is not allowed. It is allowed to use generative AI tools for formatting tasks (tables, colors on plots, creating appropriate plots legend(s), axis tick mark size, etc.).
- Include a (simple) declaration of any generative AI used. For example, "I used Claude.ai to help format the output .cvs file in Problem 42, and help with plot formatting of all figures."
- Good luck!!!

Example Cover Page (0.5 pts.)

AMAS Exam Solutions

David “Jason” Koskinen (XDN365)

Advanced Methods in Applied Statistics

August 2-3, 2042

I, Jason Koskinen, expressly vow to uphold my scientific, academic, and moral integrity by working individually on this exam and soliciting no direct external help or assistance.

AI Declaration: I used Claude.ai to help format the output .cvs files in Problem 42, and to help with the plot formatting of all figures.

Problem 1 (3.0 pts.)

- There is a file posted online which has 6 columns, each representing data of interest generated from some underlying function. There are 4121 entries, i.e. rows.
  - http://www.nbi.dk/~koskinen/Teaching/AdvancedMethodsInAppliedStatistics2026/data/Exam_2026_Prob1.txt
  - The variables/columns are independent distributions with **no** correlation to the data in the other columns
  - Be mindful about accounting for truncated ranges, as well as likelihood functions that have periodic components which will create local minima/maxima

Lists of Distributions

$$-10 \le a \le 10$$
$$-10 \le b \le 10$$
$$4000 \le c \le 8000$$

- The data in each column is produced from functions ***similar to***, or potentially exactly the same as, f(x) or f(k) shown at right
- Note that the displayed functions may be unnormalized
  - Hint: Some functions will require a normalization to convert them to probability distribution functions
  - The functions f(x) have bounds on their parameters a, b, and c

$$f(x) \propto \begin{cases} \dfrac{1}{x+5}\sin(ax) \\ \sin(ax)+1 \\ \sin(ax^2) \\ \sin(ax+1)^2 \\ x\tan(x) \\ 1+ax+bx^2 \\ 5+ax \\ \sin(ax)+ce^{bx}+1 \\ e^{-\frac{(x-a)^2}{2b^2}} \end{cases}$$

$$f(k) \propto \begin{cases} \dbinom{n}{k}p^k(1-p)^{n-k} & \textit{binomial} \\ \dfrac{\lambda^k e^{-\lambda}}{k!} & \textit{poisson} \\ \dfrac{-1}{\ln(1-p)}\dfrac{p^k}{k} & \textit{logarithmic} \end{cases}$$

Problem 1a (2.2 pts.)

- Use the separate data from first, third, and fourth columns to identify the function on the previous slide from which each was generated. Find the *best-fit values* and 68.27% *uncertainties* (i.e. 1σ confidence intervals) on those values for the distribution using a ***likelihood method*** (either bayesian or maximum likelihood is fine)
  - E.g. if f(x)=sin(ax+b)*exp(-x+c)+x/k! were one of the functions, then find the best-fit values for a, b, c, and k and their uncertainties
  - Degeneracies exist, e.g. sin(x)=cos(a+x), which can produce functionally identical data distributions
  - Provide quantitive justification that your selected function and best-fit values provide a 'reasonable' fit to the data. Any function, with associated best-fit parameters which is **statistically compatible** with the data in the files will be accepted as a proper solution. Only one solution is necessary, but needs to be **justified** as statistically compatible.
- The first and second columns have artificially truncated ranges
  - First column is only sampled in the independent variable from 6 to 11
  - Second column isn't used so don't worry about the truncation

Problem 1b (0.8 pts.)

- Plot the data and the corresponding best-fit function on the same plots
  - 3 separate 1-dimensional plots
  - Plot as a function of the independent variable
  - Make a histogram of the data, and **scale** the best-fit function to be 'reasonable' so that the features of both the data and best-fit function can be visually compared

Problem 2 (2.0 pts.)

- There is a file posted online (http://www.nbi.dk/~koskinen/Teaching/AdvancedMethodsInAppliedStatistics2026/data/Exam_2026_Problem2.txt) with data.
  - The first column is the azimuth angle of the data point
  - The second column is the zenith angle of the data point
  - There are 139 paired data points in total
  - The values are in units of radian

Problem 2a

- Correctly quantify whether the data is spherically isotropically distributed
  - Include any supporting plots, discussion, and numbers
  - A spherically isotropic distribution is uniform in the azimuth angle from 0 to 2π, and uniform in cos(zenith angle) from -1 to 1
    - Hint: you can use Monte Carlo generated pseudo-experiments to produce a test-statistic distribution of a spherically isotropic distribution.
    - Hint: isotropically distributed means 'uniform' **simultaneously** in azimuth and cos(zenith).
    - Hint: establishing that a distribution is uniform in variable 1 and separately establishing that a distribution is uniform in variable 2 **does not conclusively establish** that the 2D distribution is uniform.

Problem 2b

- Test whether the data fits the two following alternative hypotheses better than the isotropic hypothesis:
  - Hypothesis A: That 20% of the total sample is uniformly distributed in azimuth over the range {0.225π, 0.725π} and uniformly distributed in cos(zenith) over the range {-1, 0}, and the remaining 80% is fully isotropic
  - Hypothesis B: That 15% of the total sample is uniformly distributed in azimuth over the range {0π, 1π} and uniformly distributed in cos(zenith) over the range {-1, 0}, and the remaining 85% is fully isotropic.
  - Report the two p-values:
    - p-value_a : $H_{isotropic}$ versus H_A
    - p-value_b : $H_{isotropic}$ versus H_B

Problem 3 (2.0 points)

- State Highway 1 (SH1) on the South Island of New Zealand spans from the Picton Railway Station in the town of Picton (north) to Stirling Point (south).
- A file containing the population centers we will use in this problem can be found at https://www.nbi.dk/~koskinen/Teaching/AdvancedMethodsInAppliedStatistics2026/data/NZCities.txt. It includes city or town name, approximate population, distance along SH1 from Picton railway station, and region.

Problem 3a (1 point)

- Make a plot of the population estimate as a function of distance along SH1 using kernel density estimation with gaussian kernels that have a bandwidth=25km.
  - The plotted distance range (x-axis) should span from -100 km to 1025 km
  - Note: the kernel(s) should be relative to the population size. Be sure that the probability distribution function from the KDE is highest around the largest population center(s) and small far away from population centers.

- Using the KDE, what is the total population estimate from 405 km to 430 km along SH1?

Problem 3b (0.5 points)

*(https://ehinz.ac.nz/assets/Surveillance-reports/Released_2021/Number-of-Vehicles.pdf)

- Different regions of New Zealand along SH1 have varying electric car (i.e. light electric vehicle) ownership per capita, according to a 2021 report*. For distances along SH1 the following applies:
  - Nelson-Marlborough region: <91km along SH1 there are 5.5 e-cars per 1k people
  - Canterbury region: 91km - 557km there are 5.8 e-cars per 1k people
  - Otago region: 557km - 823km there are 6.0 e-cars per 1k people
  - Southland region: >823km there are 1.9 e-cars per 1k people

- Using the population KDE derived in the 3a, how many e-cars are there in each of the 4 regions along SH1?
  - Round the answers to the nearest whole number, i.e. integer values
  - Hint: you can numerically or analytically integrate the population from the population KDE and then multiply by the e-car density in that region.

Problem 3c (0.5 points)

- Show a plot of the distribution of the number of e-cars along SH1 using kernel density estimation
  - The kernel should be an epanechnikov-type with bandwidth = 35km
  - Each population center is in a region of NZ that has an estimate of the number of e-cars per 1k people. This problem is very similar to 3a, but now we want to have an estimate of the e-car distribution instead of population.

- To further refine the number of e-cars along SH1, cameras are placed at specific points along SH1 to get data about e-cars.
  - *continued on the next slide*

Problem 3c (0.5 points) continued

- The cameras are placed at the following distances along SH1 from the Picton Railway Station, and record the number of unique e-cars in 1 week. The camera data will be used to create a probability distribution function via kernel density estimation using a gaussian kernel and a bandwidth of 50 km.
  - 100 km : 12 e-cars, 200 km : 5 e-cars, 300 km : 178 e-cars, 400 km : 113 e-cars, 500 km : 42 e-cars, 600 km : 67 e-cars, 700 km : 23 e-cars, 800 km : 2 e-cars
- Make a plot of the posterior distribution where the prior comes from the KDE using the population centers and estimated e-car ownership in each region (epanechnikov KDE), and the likelihood comes from a KDE using the camera data. Be sure that the distribution for the prior **and** the PDF from the likelihood are normalized and that the posterior is also properly normalized.
- Using the posterior, what is the fraction of e-cars in New Zealand along SH1 between 400 km to 700 km?

Problem 4 (1.0 pts.)

Credit: Luca Galuzzi - www.galuzzi.it

- Data from the World Glacier Inventory* includes information from over 130k glaciers on Earth. The elevation of a subsample of 1000 glaciers is included at https://www.nbi.dk/~koskinen/Teaching/data/GlacierElevation.txt

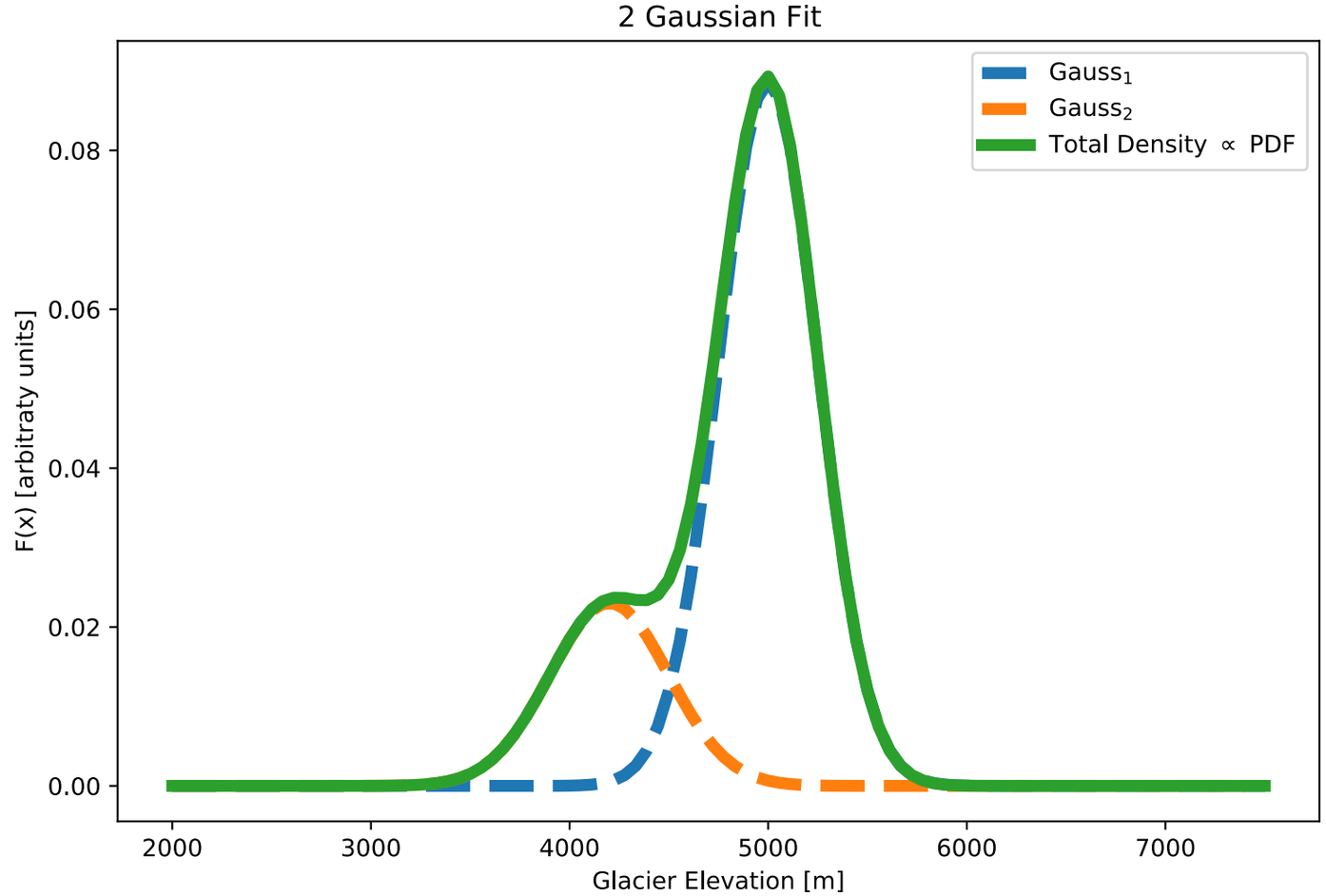
*https://nsidc.org/data/g01130/versions/1/documentation

Problem 4a (0.5 pts.)

- Fit two different hypotheses to the glacier elevation data:
  - Hypothesis A — Double gaussian fit
  - Hypothesis B — Triple gaussian fit
  - Plot the total probability distribution function for each hypothesis

Problem 4a (illustration)

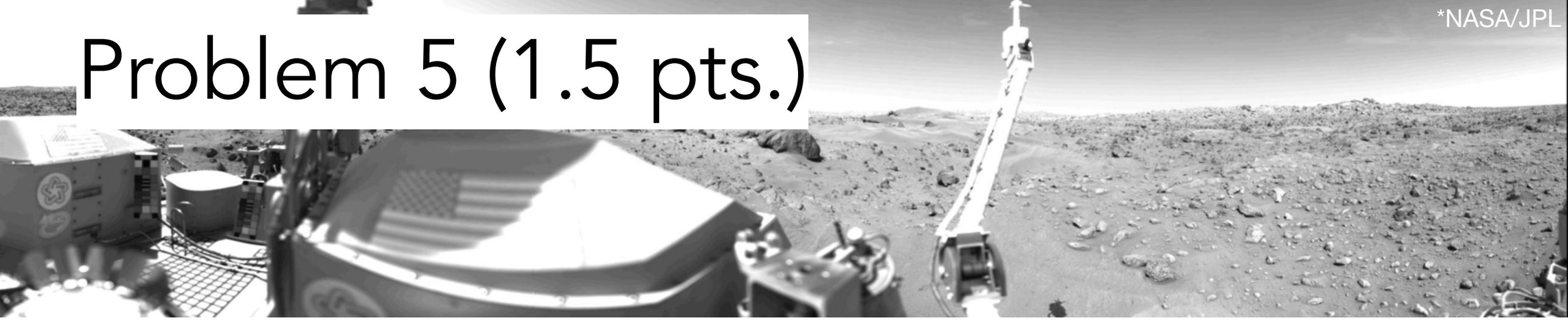
- Below is an example illustration of a double-gaussian fit which uses a function which is proportional to the actual PDF and built from 2 gaussian distributions. This illustration is not fit to any data, and is only an illustration.



Problem 4b (0.5 pts.)

- At what statistical significance can Hypothesis A be excluded/rejected when compared to Hypothesis B for the data?
  - Provide a quantitative number(s) calculated from the data, and your reasoning for the hypothesis testing acceptance/rejection.
  - Hint: a p-value is a useful way to report hypothesis testing results.

Problem 5 (1.5 pts.)



- The success of planetary exploration on Mars is highly dependent on the ability of any exploratory vehicle, e.g. Mars Perseverance rover and Ingenuity helicopter, to survive the low temperatures as well as the temperature changes.
- Viking 1 was the first successful Mars landing and took temperature data that was used to inform future Mars missions.
- We will use temperatures recorded at different time intervals from Viking 1
  - The first entry is the sol (the Mars analog of an Earth day) and the second is the temperature in Celsius. We will assume the temperature uncertainty is ± 0.001C.
  - For example [203.41, -89.37] is data taken on sol 203.41 and the temperature is -89.37 C.

```
array([[ 203.41 ,  -89.37 ],
       [ 203.435,  -94.88 ],
       [ 203.46 , -101.25 ],
       [ 203.484, -106.52 ],
       [ 203.509, -108.66 ],
       [ 203.534, -114.25 ],
       [ 203.558, -114.30 ],
       [ 203.583, -117.66 ],
       [ 203.608, -122.45 ]])
```

Problem 5a (0.5 points)

- Use the data provided on the previous page to create 2 splines:
  - A linear spline
  - A cubic spline **<u>or</u>** a Piecewise Cubic Hermite Interpolating Polynomial (PCHIP) spline. Create one of these non-linear splines, but not both.
- What are the estimates of the temperature on sol 203.570 from the created linear-spline, as well as the created cubic/PCHIP-spline?

Problem 5b (0.5 pts.)

- Make a scatter plot of temperature versus time of the interpolated temperatures from the linear-spline as well as the cubic/PCHIP-spline covering the time range from 203.410 sol to 203.608 sol.
  - To ensure we can see any interesting features of the linear and cubic/PCHIP splines, make sure there are at least 200 points for each interpolation.

- During sol 203.410 to 203.608, if we know that the temperature should be continuously dropping, are there any regions of time that we should look at more closely for the cubic/PCHIP spline to make sure that the interpolated temperature is monotonically decreasing?

Problem 5c (0.5 points)

- Imagine that proposed electronics components for a Mars rover, or other Mars planetary-surface exploration vehicle, are unable to sustain temperature changes of more than 0.09 C within 0.0004 sol.
- Would new electronics be needed with more robustness to temperature fluctuations according to your interpolation(s) to the Viking 1 data?